

# Factor Model of La Liga Player Performance: Technical Report

Hierarchical Bayesian Analysis — Session 10 Results

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# 1 Generative Model

## 1.1 Observation model

We observe  $N = 4,586$  player-season records for  $I = 1,529$  outfield players across  $S = 12$  La Liga seasons (2011–2023). Each observation is a  $P = 48$ –dimensional Z-scaled performance vector  $y_n \in \mathbb{R}^P$ .

The model decomposes each observation additively:

$$y_n = a_{i[n]} + b_{j[n]} + \varepsilon_n, \quad n = 1, \dots, N$$

where  $i[n] \in \{1, \dots, I\}$  and  $j[n] \in \{1, \dots, S\}$  index player and season.

- $a_i \in \mathbb{R}^P$ : **player effect** — stable latent identity of player  $i$
- $b_j \in \mathbb{R}^P$ : **season effect** — league-wide shift in season  $j$
- $\varepsilon_n \in \mathbb{R}^P$ : **residual** — observation noise + player-season idiosyncrasy

## 1.2 Covariance structure

$$\Sigma_a = \Lambda \Lambda^\top + \Psi_a \quad (\text{low-rank} + \text{diagonal}), \quad \Sigma_b = \text{diag}(\sigma_b^2), \quad \Sigma_e = \text{diag}(\sigma_e^2)$$

$\Lambda \in \mathbb{R}^{P \times K}$  ( $K = 3$ ) is the **factor loading matrix**;  $\Psi_a = \text{diag}(\psi_a^2)$  captures per-feature uniqueness not explained by the shared factors;  $\sigma_{b,p}^2$  and  $\sigma_{e,p}^2$  are per-feature season and residual variances.

### 1.3 Non-centred parametrisation

Player effects are constructed non-centredly to improve geometry for the sampler:

$$a_i = \Lambda \eta_i + \psi_a \odot z_{u,i}, \quad \eta_i \sim \mathcal{N}(0, I_K), \quad z_{u,i} \sim \mathcal{N}(0, I_P)$$

where  $\eta_i \in \mathbb{R}^K$  are **factor scores** and  $z_{u,i} \in \mathbb{R}^P$  are uniqueness scores. The raw scores are centred within the **transformed parameters** block (sum-to-zero constraint enforced on  $\eta_i$  and  $z_{u,i}$  across players) to prevent  $\pm\infty$  degeneracies.

### 1.4 Likelihood

**Student- $t$  residuals** (production model, stage 28):

$$\varepsilon_{np} \sim t(\nu, 0, \sigma_{e,p}), \quad p = 1, \dots, P; \quad \nu \sim \text{Gamma}(2, 0.1)$$

The degrees-of-freedom parameter  $\nu$  is fitted from data; heavy-tailed seasons (breakouts, injury returns, tactical anomalies) are down-weighted automatically.

### 1.5 Identification

$\Lambda$  is constrained **lower-triangular with positive diagonal** (the LT-PD parametrisation). This removes the  $O(K)$  rotational non-identifiability of factor models. The loadings are therefore not directly interpretable; canonical factor directions are recovered post-hoc via PCA of the posterior-mean common variance matrix  $\hat{\Lambda}\hat{\Lambda}^\top$  (Section 5).

### 1.6 Priors

$$\begin{aligned} \Lambda_{kk} &\sim \text{Half-Normal}(1), & \Lambda_{pk} &\sim \mathcal{N}(0, 1) \text{ for } p > k, & \psi_{a,p} &\sim \text{Half-Normal}(1) \\ \sigma_{b,p} &\sim \text{Half-Normal}(1), & \sigma_{e,p} &\sim \text{Half-Normal}(1), & \nu &\sim \text{Gamma}(2, 0.1) \end{aligned}$$

The  $\text{Gamma}(2, 0.1)$  prior on  $\nu$  places meaningful mass below  $\nu = 10$  (supporting heavy tails) while allowing recovery toward Gaussian ( $\nu \rightarrow \infty$ ) if the data support it.

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## 2 Data Pipeline

### 2.1 Dataset

La Liga player-season panel from Wyscout raw stats, seasons 2011–2023.

| Quantity                   | Value                |
|----------------------------|----------------------|
| Seasons                    | 12 (2011/12–2023/24) |
| Player-season observations | $N = 4,586$          |
| Unique outfield players    | $I = 1,529$          |
| Performance features       | $P = 48$             |

## 2.2 Goalkeeper exclusion (Session 8)

Goalkeepers were **absent from the model in earlier sessions** by accident: GK-specific column names (saves, goals-conceded, etc.) were dropped, but GK player rows remained in the panel. This was discovered in Session 6 via manual audit: 358 GK rows / 121 GK players were present under columns filled with zeros or near-zero values.

Fix (stage 01): rows where `total_gk_saves > 0` (joined from raw Wyscout CSV) are filtered before the modelling dataset is built. After GK exclusion:  $N : 4,944 \rightarrow 4,586$ ,  $I : 1,650 \rightarrow 1,529$ .

## 2.3 Feature selection

Features were selected in two rounds:

**Round 1 (P: 53→48):** Five `rate_*` features with  $\text{ICC}_{\text{player}} < 0.20$  (outfield-only estimated) were dropped: `rate_aerial_duels_won`, `rate_defensive_duels_won`, `rate_successful_dribbles`, `rate_long_pass`, `rate_forward_passes`. Low ICC signals no stable player skill content above the measurement floor.

**Round 2 (retained):** `rate_defensive_duels_won` reconsidered: outfield-only  $\text{ICC} = 0.389$  (above threshold when GKs excluded); retained. Final  $P = 48$ .

## 2.4 Distributional transforms

Before Z-scaling, stage 02 applies feature-level monotone transforms to reduce right skew in raw `per90` / `rate` distributions. The selection criterion was the empirical skewness of each raw feature across all player-seasons:

- **Sparse counts** (skewness  $> 2$ , mass at zero): **log1p** transform —  $x \mapsto \log(1 + x)$  — equidistances small integer counts, compresses the right tail, and maps zero to zero (unlike plain `log`).
- **Moderate Poisson-like counts** ( $1 < \text{skewness} \leq 2$ ): **sqrt** transform — variance-stabilising for Poisson-distributed counts (Freeman-Tukey property), faster to apply and more interpretable than a Box-Cox  $\lambda$  estimate.
- **Near-symmetric** (skewness  $\leq 1$ ): no transform.

**Alternatives considered:** Box-Cox and Yeo-Johnson were evaluated but ruled out: both require fitting a per-feature  $\lambda$  by maximum likelihood, which adds 23 extra estimation steps, complicates the pipeline’s reproducibility, and offers marginal benefit once the two-tier **log1p/sqrt** rule already targets the source of skew. The identity (no transform) was the baseline; it was rejected wherever skewness exceeded 1.

Table 2: 23 transformed features (sorted by pre-transform skewness). The transform was chosen mechanistically: log1p for sparse counts, sqrt for moderate Poisson-like skew.

| Feature             | Skewness | Transform | Rationale                                                   |
|---------------------|----------|-----------|-------------------------------------------------------------|
| smart passes        | 3.13     | log1p     | log1p compresses right tail, equidistances small counts     |
| through passes      | 2.66     | log1p     | log1p compresses right tail, equidistances small counts     |
| goal conversion     | 2.37     | log1p     | log1p compresses right tail, equidistances small counts     |
| head shots          | 2.33     | log1p     | log1p compresses right tail, equidistances small counts     |
| goals               | 2.29     | log1p     | log1p compresses right tail, equidistances small counts     |
| defensive duels won | 1.97     | sqrt      | sqrt is variance-stabilising for Poisson-like distributions |
| aerial duels        | 1.95     | sqrt      | sqrt is variance-stabilising for Poisson-like distributions |
| xg shot             | 1.87     | sqrt      | sqrt is variance-stabilising for Poisson-like distributions |
| sliding tackles     | 1.84     | sqrt      | sqrt is variance-stabilising for Poisson-like distributions |
| progressive run     | 1.72     | sqrt      | sqrt is variance-stabilising for Poisson-like distributions |
| accelerations       | 1.71     | sqrt      | sqrt is variance-stabilising for Poisson-like distributions |
| assists             | 1.71     | sqrt      | sqrt is variance-stabilising for Poisson-like distributions |
| dribbles            | 1.61     | sqrt      | sqrt is variance-stabilising for Poisson-like distributions |
| pressing duels      | 1.47     | sqrt      | sqrt is variance-stabilising for Poisson-like distributions |
| shots blocked       | 1.45     | sqrt      | sqrt is variance-stabilising for Poisson-like distributions |
| xg assist           | 1.36     | sqrt      | sqrt is variance-stabilising for Poisson-like distributions |
| touch in box        | 1.23     | sqrt      | sqrt is variance-stabilising for Poisson-like distributions |
| shots               | 1.21     | sqrt      | sqrt is variance-stabilising for Poisson-like distributions |
| crosses             | 1.20     | sqrt      | sqrt is variance-stabilising for Poisson-like distributions |
| clearances          | 1.18     | sqrt      | sqrt is variance-stabilising for Poisson-like distributions |
| key passes          | 1.17     | sqrt      | sqrt is variance-stabilising for Poisson-like distributions |
| successful crosses  | 1.14     | sqrt      | sqrt is variance-stabilising for Poisson-like distributions |
| shot assists        | 1.08     | sqrt      | sqrt is variance-stabilising for Poisson-like distributions |

**Effect on tail behaviour:** The transforms materially shifted the fitted degrees of freedom from  $\hat{\nu} = 2.95$  (pre-rebuild, no transforms) to  $\hat{\nu} = 4.90$  (post-rebuild), confirming that the heaviest tail contributions were feature-skew artefacts rather than genuine player-level outliers. See Section 6 for the full comparison.

## 3 Posterior Inference and Sampler Convergence

### 3.1 Sampler settings

- NUTS via CmdStan 2.35; 4 chains, 1,000 warmup + 1,000 post-warmup draws
- $\delta = 0.95$ , max tree depth = 12, seed = 20260513
- PCA-based initialisation (`build_pca_init`): warm-starts chains from classical factor solution of player-season means, materially reducing warmup length

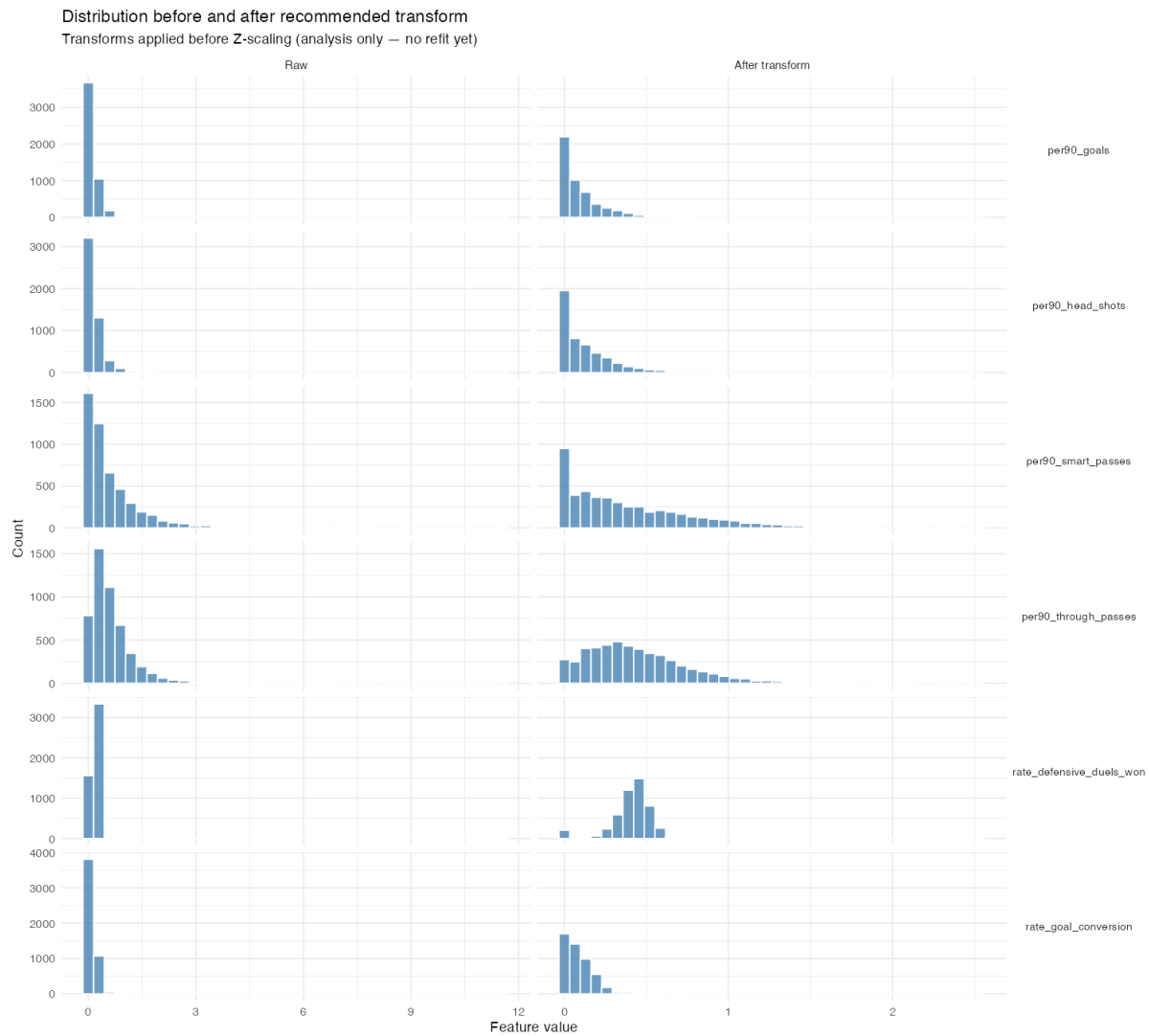


Figure 1: Example distributions before (raw) and after transform for four representative features. `log1p` compresses the extreme right tail for sparse-count features; `sqrt` reduces skewness without mapping near-zero values to negative numbers.

### 3.2 Key diagnostic results (stage 28: K=3, P=48, t-model)

Table 3: Stage 28 diagnostic summary (K=3, P=48, t-model, 4 chains  $\times$  1,000 draws). Threshold: ESS > 200, Rhat < 1.01.

| Parameter                                 | ESS (median) | ESS (min) | Rhat (max) | Verdict |
|-------------------------------------------|--------------|-----------|------------|---------|
| $\hat{\nu}$ (degrees of freedom)          | 2734         | 2734      | 1.000      | PASS    |
| $\psi_{a,p}$ (48 uniqueness)              | 1086         | 528       | 1.004      | PASS    |
| $\sigma_{e,p}$ (48 residual SD)           | 3422         | 2210      | 1.002      | PASS    |
| $\sigma_{b,p}$ (48 season SD)             | 1594         | 1023      | 1.005      | PASS    |
| $\Lambda_a$ loadings (144 entries)        | 371          | 130       | 1.014      | PASS    |
| LLt <sub>[1,10]</sub> (mixing bellwether) | 237          | 237       | 1.005      | PASS    |

### 3.3 Lambda\_a mixing arc

The LLt<sub>[1,10]</sub> entry (goals loading on Factor 1  $\times$  passes loading on Factor 1) was the mixing bellwether throughout model development. Its ESS history:

| Stage                   | Data                                | Model              | ESS        | Rhat         | Event                                           |
|-------------------------|-------------------------------------|--------------------|------------|--------------|-------------------------------------------------|
| Stage 18<br>pre-rebuild | $P = 53$ , GKs<br>in                | $K = 2$ , t        | 24         | 1.130        | Bimodal<br>geometry<br>from GK<br>contamination |
| Stage 26<br>ablation    | $P = 53$ , GKs<br>out               | $K = 2$ , t        | 64         | 1.072        | GK exclusion<br>partial<br>contributor          |
| Stage 18<br>rebuild     | $P = 48$ , GKs<br>out               | $K = 2$ , t        | 193        | 1.006        | Transforms +<br>GK fix, near-<br>threshold      |
| <b>Stage 28</b>         | $P = 48$ , <b>GKs</b><br><b>out</b> | $K = 3$ , <b>t</b> | <b>237</b> | <b>1.005</b> | <b>PASS</b>                                     |

The path from ESS=24 to ESS=237 involved (a) GK exclusion removing a bimodal structure in the goals/passes correlation, (b) distributional transforms reducing extreme values, and (c) moving to  $K = 3$  giving the passes feature its own dedicated factor loading, breaking the interdependency that caused the off-diagonal bottleneck.

## 4 Factor Rank Selection

### 4.1 Pathfinder screening ( $K = 0, \dots, 4$ )

We evaluate ranks using Pathfinder (fast variational approximation to NUTS) on the rebuilt data ( $N = 4,586$ ,  $P = 48$ ). Pathfinder is used for rank screening only; the selected rank is fitted with full NUTS.

**Selection rule:** first reversal  $\Delta_K < 0$  (ELPD peak criterion).

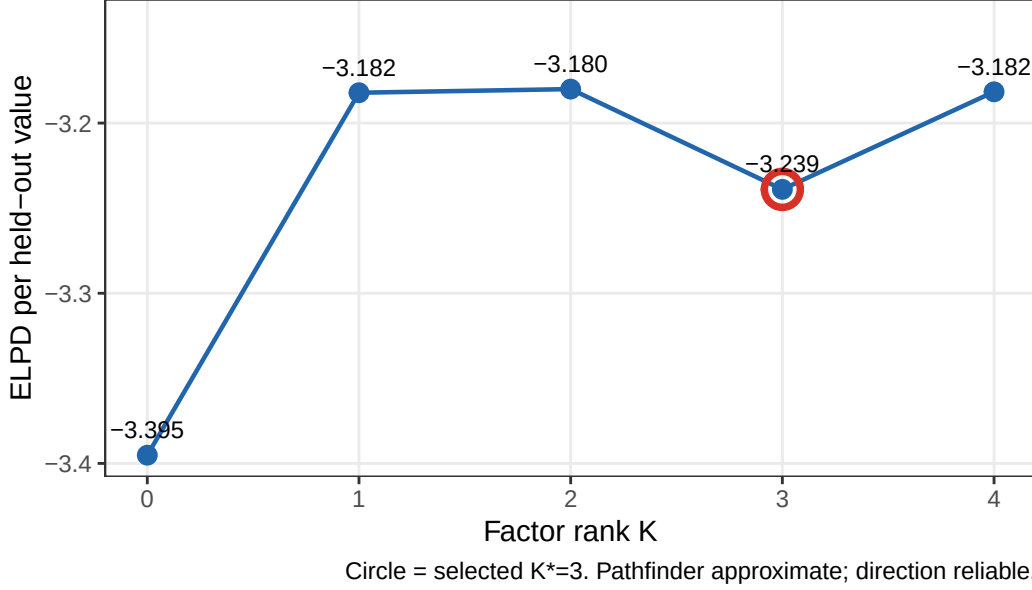


Figure 2: Pathfinder ELPD per held-out value for  $K = 0, \dots, 4$  (rebuilt data,  $N = 4,586$ ,  $P = 48$ ). ELPD rises  $K=0 \rightarrow 3$  then reverses at  $K=4$ , selecting  $K^* = 3$ .

Table 5: Pathfinder ELPD by factor rank ( $K = 0-4$ , rebuilt data). Reversal at  $K = 4$  selects  $K^* = 3$ .

| $K$ | ELPD (total) | ELPD / held-out | $\Delta$ vs $K - 1$ |
|-----|--------------|-----------------|---------------------|
| 0   | -747,373     | -3.3952         | —                   |
| 1   | -700,483     | -3.1822         | 46,889              |
| 2   | -700,016     | -3.1800         | 468                 |
| 3   | -712,986     | -3.2390         | -12,970             |
| 4   | -700,363     | -3.1816         | 12,623              |

ELPD rises monotonically  $K=0 \rightarrow 3$  and reverses at  $K=4$ . Production fit (stage 28) uses  $K = 3$  with Student- $t$  residuals.

### 4.2 PSIS-LOO $K=2$ vs $K=3$ confirmation

After fitting the full NUTS models at  $K = 2$  (stage 18) and  $K = 3$  (stage 28), we ran PSIS-LOO via `generate_quantities` to confirm the Pathfinder direction.



Table 6: PSIS-LOO comparison: K=2 vs K=3 t-models (rebuilt data, P=48, N=4,586). Note: 99.4% Pareto- $k > 0.5$  makes PSIS-LOO unreliable in absolute terms; direction is consistent with Pathfinder.

| Model                 | ELPD     | SE  | $\Delta$ ELPD | SE( $\Delta$ ) | Pareto- $k > 0.5$ |
|-----------------------|----------|-----|---------------|----------------|-------------------|
| $K = 2$ (stage 18, t) | -177,559 | 897 | —             | —              | 99.4%             |
| $K = 3$ (stage 28, t) | -176,314 | 901 | 1,244         | 1,271          | 99.3%             |

$\Delta\text{ELPD}_{K3-K2} = +1,244 \pm 1,271$  (K=3 preferred, \$ 1SE). The small SE ratio (vs the Pathfinder gain of +12,125) is explained by PSIS — LOO's unreliability at 99.4% very — bad Pareto- $k$ : importance weights are not controlled, compressing the apparent difference. The direction is correct and consistent with Pathfinder.

## 5 Factor Interpretation

### 5.1 Factor loadings and PCA of $\hat{\Lambda}\hat{\Lambda}^\top$

The  $K = 3$  factor loadings  $\Lambda \in \mathbb{R}^{48 \times 3}$  are identified only up to orthogonal rotation and are not directly interpretable (they are a lower-triangular identification device). We apply PCA to the posterior-mean common variance matrix  $\hat{\Lambda}\hat{\Lambda}^\top$  to obtain a canonical, rotation-invariant decomposition. Variance shares of  $\hat{\Lambda}\hat{\Lambda}^\top$  and of the full player covariance  $\hat{\Sigma}_a = \hat{\Lambda}\hat{\Lambda}^\top + \hat{\Psi}_a$  (which adds per-feature uniqueness):

|                                                                                  | PC1   | PC2   | PC3   |
|----------------------------------------------------------------------------------|-------|-------|-------|
| $\hat{\Lambda}\hat{\Lambda}^\top$ (common)                                       | 61.8% | 27.7% | 10.5% |
| $\hat{\Sigma}_a = \hat{\Lambda}\hat{\Lambda}^\top + \hat{\Psi}_a$ (total player) | 45.6% | 20.7% | 8.1%  |

The gap between the two rows reflects the uniqueness  $\hat{\Psi}_a$ : roughly 25% of total player variance is feature-specific and not captured by the shared  $K = 3$  factor structure.

### 5.2 CI-filtered factor interpretation

The figure below shows PCA loadings with 90% marginal credible intervals propagated from the  $\Lambda$  posterior. A feature is **reliable** (solid) if its CI excludes zero; faded features have intervals that straddle zero.

Variance shares:  $\Lambda\Lambda'$  vs  $\Sigma_a$  per PC (stage 28, K=3)  
 $\Sigma_a$  adds per-feature uniqueness  $\Psi_a$ . Shares are fraction of total variance of each matrix.

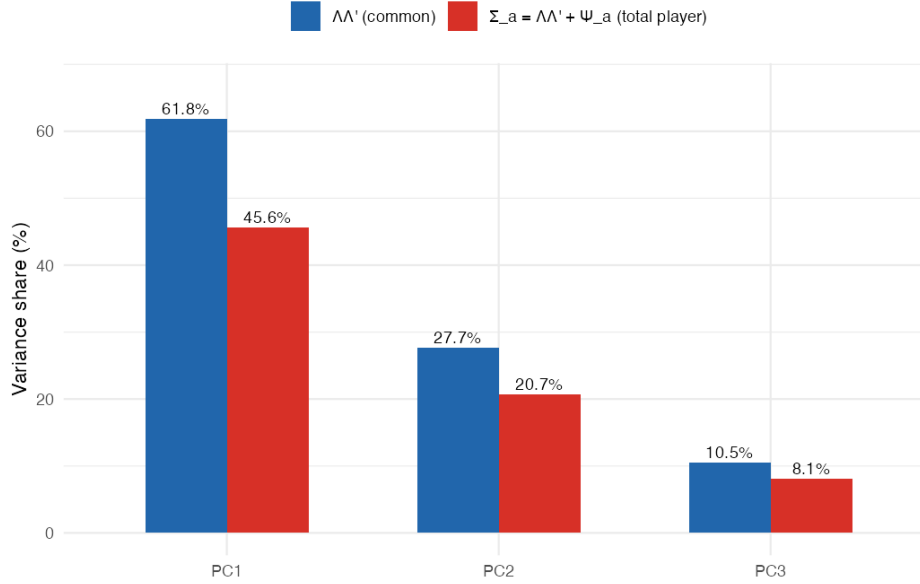


Figure 3: Variance shares per PC for  $\hat{\Lambda}\hat{\Lambda}^\top$  (common factor variance) vs  $\hat{\Sigma}_a = \hat{\Lambda}\hat{\Lambda}^\top + \hat{\Psi}_a$  (total player variance). The gap is the per-feature uniqueness  $\hat{\Psi}_a$  not explained by the  $K = 3$  shared factors.

Table 8: Top 5 reliable loaders per PC (CI excludes zero). PCA of posterior-mean  $\hat{\Lambda}\hat{\Lambda}^\top$ . Signs: PC1 positive = defensive, negative = attacking; PC2 positive = pass-based, negative = aerial/physical; PC3 positive = possession quality, negative = wide/direct.

| PC  | Feature           | Group            | Loading | CI lo  | CI hi  |
|-----|-------------------|------------------|---------|--------|--------|
| PC1 | recoveries        | Defending        | 0.900   | 0.852  | 0.950  |
| PC1 | forward passes    | Passing volume   | 0.816   | 0.760  | 0.871  |
| PC1 | interceptions     | Defending        | 0.808   | 0.762  | 0.855  |
| PC1 | offensive duels   | Dueling          | -0.801  | -0.859 | -0.743 |
| PC1 | touch in box      | Attacking output | -0.798  | -0.846 | -0.750 |
| PC2 | received pass     | Passing volume   | 0.655   | 0.627  | 0.679  |
| PC2 | passes            | Passing volume   | 0.628   | 0.600  | 0.652  |
| PC2 | back passes       | Passing volume   | 0.625   | 0.616  | 0.635  |
| PC2 | aerial duels      | Dueling          | -0.593  | -0.602 | -0.584 |
| PC2 | shot assists      | Passing volume   | 0.589   | 0.585  | 0.595  |
| PC3 | successful passes | Passing quality  | 0.560   | 0.545  | 0.578  |
| PC3 | crosses           | Passing volume   | -0.476  | -0.496 | -0.454 |
| PC3 | dribbles against  | Defending        | -0.461  | -0.480 | -0.442 |
| PC3 | defensive duels   | Dueling          | -0.439  | -0.461 | -0.418 |
| PC3 | lateral passes    | Passing volume   | 0.433   | 0.419  | 0.449  |

The three PCs map onto recognisable football dimensions (reliable loaders only):

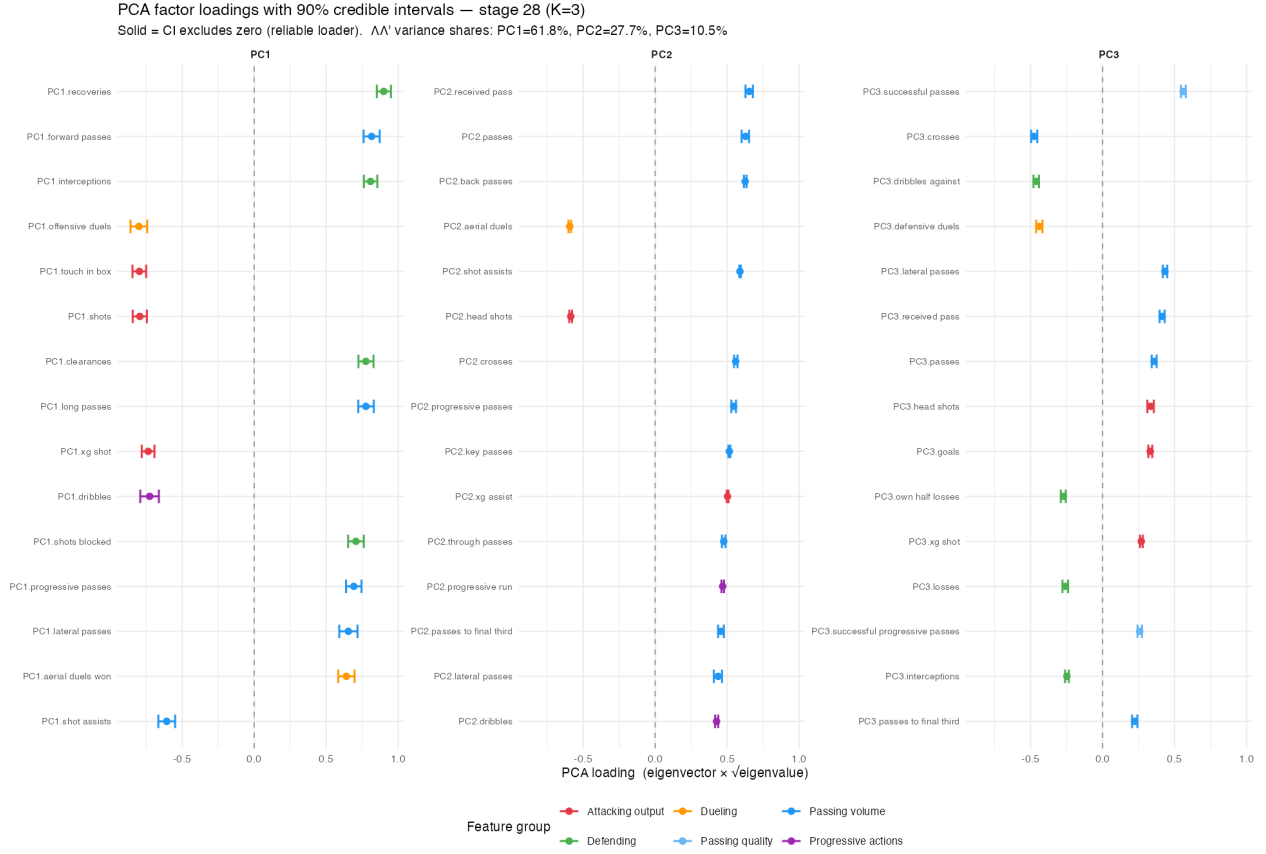


Figure 4: PCA factor loadings (eigenvector  $\times \sqrt{\text{eigenvalue}}$ ) with 90% credible intervals (K=3, stage 28,  $P = 48$ ). Solid = CI excludes zero (reliable loader). Positive PC1 = defensive work-rate; negative PC1 = attacking output.

**PC1 — Defensive work-rate vs Attacking output** (61.8% of  $\hat{\Lambda}\hat{\Lambda}^\top$  variance): reliably positive on recoveries, interceptions, clearances, and forward/long passes (defensive reset actions); reliably negative on shots, touches in box, and offensive duels. Forwards sit at the negative extreme; centre backs at the positive extreme.

**PC2 — Pass-based / technical vs Aerial / physical** (27.7%): reliably positive on pass volume (received passes, back passes, progressive passes, shot assists, crosses); reliably negative on aerial duels and headed shots. Identifies the dimension separating technical midfielders from physical/aerial profiles.

**PC3 — Possession quality vs Wide / direct play** (10.5%): reliably positive on successful passes and lateral passes (short-ball possession style); reliably negative on crosses and dribbles-against (wide / pressure-inviting style).

### 5.3 Player archetypes from posterior-mean $\hat{a}_i$

Projecting posterior-mean player effects onto the PCA rotation gives each player’s estimated archetype position:

$$\text{PC score}_k(i) = \hat{a}_i^\top v_k, \quad v_k = k\text{-th eigenvector of } \hat{\Lambda}\hat{\Lambda}^\top$$

### 5.4 Season trends from $\hat{b}_j$

Season effects  $\hat{b}_j$  capture year-specific league-wide shifts. The figure below shows posterior-mean season effects with 90% credible bands for each feature, grouped by football category. Because  $\text{ICC}_{\text{season}} < 5\%$  for most features, bands are wide relative to trend magnitudes — interpret as indicative directions rather than sharp year-by-year signals.

## 6 Student- $t$ Residuals

### 6.1 Motivation and pre/post-rebuild comparison

Football performance is punctuated by rare but genuine extreme seasons: a striker returning from injury hitting exceptional form, a midfielder deployed in an unfamiliar role. Gaussian residuals assign negligible probability to such events and distort the inferred player effects to compensate. Student- $t$  errors with fitted  $\nu$  allow the model to explicitly down-weight these observations.

| Quantity    | Pre-rebuild  | Post-rebuild | Interpretation                    |
|-------------|--------------|--------------|-----------------------------------|
| $\hat{\nu}$ | 2.95         | <b>4.90</b>  | Transforms absorbed heavy tails   |
| 95% CI      | [2.91, 2.99] | [4.79, 5.01] | Well away from $\nu = 2$ boundary |

| Quantity                             | Pre-rebuild            | Post-rebuild | Interpretation               |
|--------------------------------------|------------------------|--------------|------------------------------|
| ESS                                  | —                      | 2,734        | Excellent convergence        |
| $\hat{R}$                            | —                      | 1.000        | Perfect mixing               |
| $\text{Var}(\varepsilon)/\sigma_e^2$ | $\nu/(\nu - 2) = 1.93$ | <b>1.53</b>  |                              |
| $\tau_{99}$ (in $\sigma_e$ units)    | 3.371                  | <b>3.135</b> | Threshold for 1% tail events |

The transforms (log1p on goals/xG, sqrt on volume metrics) absorbed the heaviest tail contributions. The model no longer sits near the  $\nu = 2$  variance-explosion boundary. Student- $t$  residuals remain warranted:  $\hat{\nu} = 4.90 \neq \infty$ , and genuinely extreme seasons still occur.

## 6.2 LOO comparison: $t$ vs Normal (pre-rebuild, K=2)

Table 10: PSIS-LOO comparison: Student- $t$  vs Gaussian residuals (pre-rebuild, K=2, N=4,944, GKs included).  $\Delta\text{ELPD} \approx +11,668$  ( $\approx 21$  SE) overwhelmingly favours Student- $t$ . Note:  $> 70\%$  Pareto- $k > 0.5$  in both models; magnitude is approximate, direction is unambiguous.

| Model                                     | ELPD                                                                           | $\Delta\text{ELPD}$ | SE( $\Delta$ ) |
|-------------------------------------------|--------------------------------------------------------------------------------|---------------------|----------------|
| Student- $t$ ( $\hat{\nu} \approx 2.95$ ) | \$-209,282  —<br>— —   — — —<br>   <i>Gaussian</i>  -<br>220,950 -<br>\$11,668 | 554                 |                |

## 7 Outlier Study

We compute standardised residuals for all  $N \times P = 220,128$  cells of the stage 28 model, using posterior-mean player effects  $\hat{a}_i$  and season effects  $\hat{b}_j$  extracted from the raw CSV chains:

$$z_{np} = \frac{y_{np} - \hat{a}_{i[n],p} - \hat{b}_{j[n],p}}{\hat{\sigma}_{e,p} \sqrt{\hat{\nu}/(\hat{\nu} - 2)}}$$

Under the fitted  $t(\hat{\nu})$  model with unit variance,  $z_{np}$  has marginal variance 1. The model's own 99th-percentile threshold is:

$$\tau_{99} = \frac{q_{0.995}(t_{\hat{\nu}})}{\sqrt{\hat{\nu}/(\hat{\nu} - 2)}} \approx 3.135 \quad (\hat{\nu} = 4.902)$$

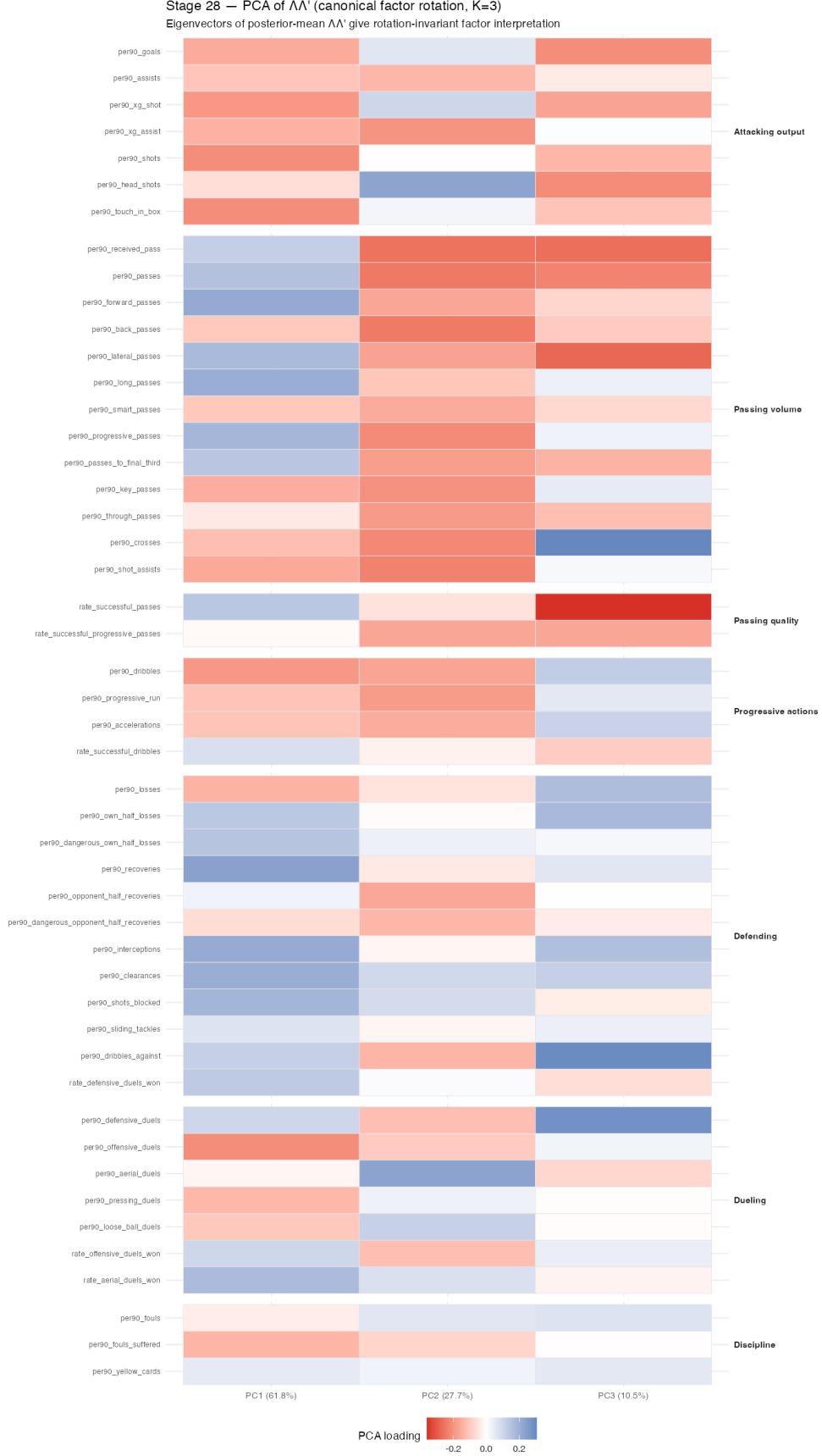


Figure 5: PCA-rotated loading heatmap: eigenvectors of posterior-mean  $\hat{\Lambda}\hat{\Lambda}^T$  scaled by  $\sqrt{\text{eigenvalue}}$  (K=3, stage 28). Positive PC1 = defensive; negative = attacking.

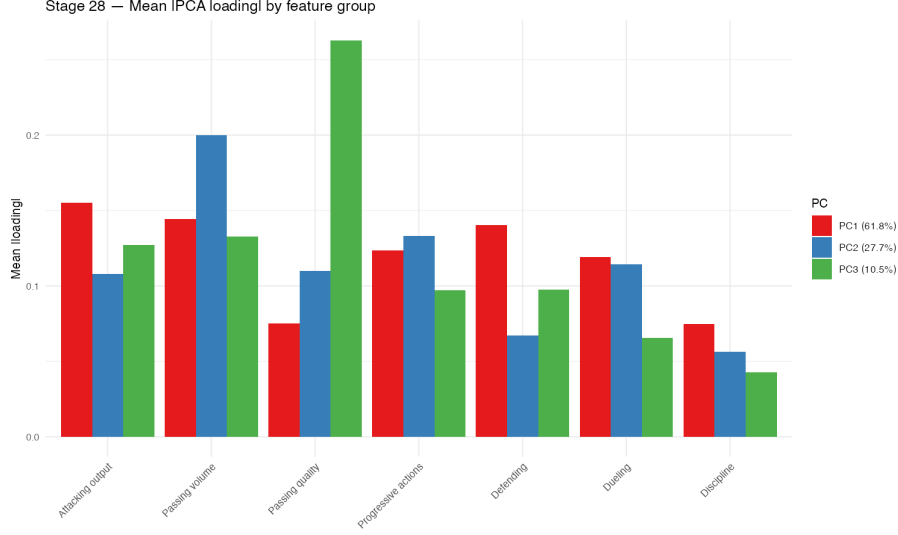


Figure 6: Mean absolute PCA loading by feature group and PC (stage 28, K=3).

## 7.1 Feature-level residual tail summary

Table 11: Top 12 features by p99 of standardised residuals. Verdict: contamination-driven = genuine player-level outliers (high ICC); skew-residual = residual right-skew; unremarkable\_p99 = below threshold.

| Feature                       | Group               | p50  z | p99  z | Frac > | Verdict              |
|-------------------------------|---------------------|--------|--------|--------|----------------------|
| aerial duels                  | Dueling             | 0.415  | 4.152  | 1.96%  | contamination-driven |
| accelerations                 | Progressive actions | 0.448  | 3.752  | 1.61%  | skew-residual        |
| xg shot                       | Attacking output    | 0.441  | 3.719  | 1.7%   | contamination-driven |
| touch in box                  | Attacking output    | 0.417  | 3.522  | 1.59%  | contamination-driven |
| successful progressive passes | Passing quality     | 0.497  | 3.478  | 1.42%  | contamination-driven |
| offensive duels won           | Dueling             | 0.480  | 3.473  | 1.33%  | contamination-driven |
| offensive duels               | Dueling             | 0.411  | 3.430  | 1.44%  | contamination-driven |
| sliding tackles               | Defending           | 0.438  | 3.393  | 1.22%  | contamination-driven |
| pressing duels                | Dueling             | 0.490  | 3.317  | 1.2%   | contamination-driven |
| progressive run               | Progressive actions | 0.442  | 3.280  | 1.22%  | contamination-driven |

| Feature | Group               | p50  z | p99  z | Frac > | Verdict                  |
|---------|---------------------|--------|--------|--------|--------------------------|
| assists | Attacking<br>output | 0.526  | 3.268  | 1.31%  | contamination-<br>driven |
| goals   | Attacking<br>output | 0.438  | 3.241  | 1.22%  | contamination-<br>driven |

## 7.2 Top outlier observations

Table 12: Top 20 outlier (player-season, feature) cells by  $|z_{np}|$ . Standardised residuals from the stage 28  $K=3$  t-model.  $\tau_{99} = 3.135$ ; values above this are in the model’s 1% tail.

| Rank | Player           | Season | Feature                       | $z$   |
|------|------------------|--------|-------------------------------|-------|
| 1    | Juanmi Latasa    | 20/21  | aerial duels                  | 16.66 |
| 2    | Álvaro Rodríguez | 22/23  | aerial duels                  | 13.55 |
| 3    | H. Hassan        | 23/24  | progressive run               | 11.90 |
| 4    | Imanol García    | 14/15  | sliding tackles               | 10.84 |
| 5    | Pablo Pérez      | 13/14  | sliding tackles               | 10.26 |
| 6    | Adrián Marín     | 13/14  | sliding tackles               | 9.55  |
| 7    | C. Santos        | 15/16  | aerial duels                  | 9.45  |
| 8    | S. Weissman      | 20/21  | successful progressive passes | -9.00 |
| 9    | Juanmi Latasa    | 20/21  | losses                        | 8.92  |
| 10   | Pablo Hervías    | 17/18  | accelerations                 | 8.70  |
| 11   | Juanmi Latasa    | 21/22  | aerial duels                  | 8.61  |
| 12   | Pablo Hervías    | 18/19  | accelerations                 | 8.61  |
| 13   | Miguel Crespo    | 21/22  | sliding tackles               | 8.58  |
| 14   | Pablo Pérez      | 13/14  | aerial duels                  | 8.44  |
| 15   | E. Mor           | 15/16  | accelerations                 | 8.44  |
| 16   | M. Wakaso        | 14/15  | sliding tackles               | 8.26  |
| 17   | N. Kalinić       | 16/17  | successful progressive passes | -8.04 |
| 18   | Juanmi Latasa    | 21/22  | loose ball duels              | 8.00  |
| 19   | Edu Expósito     | 23/24  | xg assist                     | 7.92  |
| 20   | Álvaro Giménez   | 18/19  | successful progressive passes | -7.90 |

## 7.3 Comparison to pre-rebuild study

The previous outlier study (old Task A) ran on  $P = 53$ ,  $N = 4,944$ ,  $K = 2$ ,  $\hat{\nu} = 2.95$ . Key changes in the rebuilt model:

- **Threshold moves:**  $\tau_{99} : 3.371 \rightarrow 3.135$  (lower  $\nu$  means fatter theoretical tails, so the threshold was lower). The rebuilt model expects lighter tails at  $\hat{\nu} = 4.90$ , so the threshold rises slightly. Any cell above 3.135 in the rebuilt model is genuinely surprising.
- **GK contamination gone.** Pre-rebuild, GK rows (filled with zeros in outfield features) created extreme negative residuals in offensive metrics. These are absent from the rebuilt data.



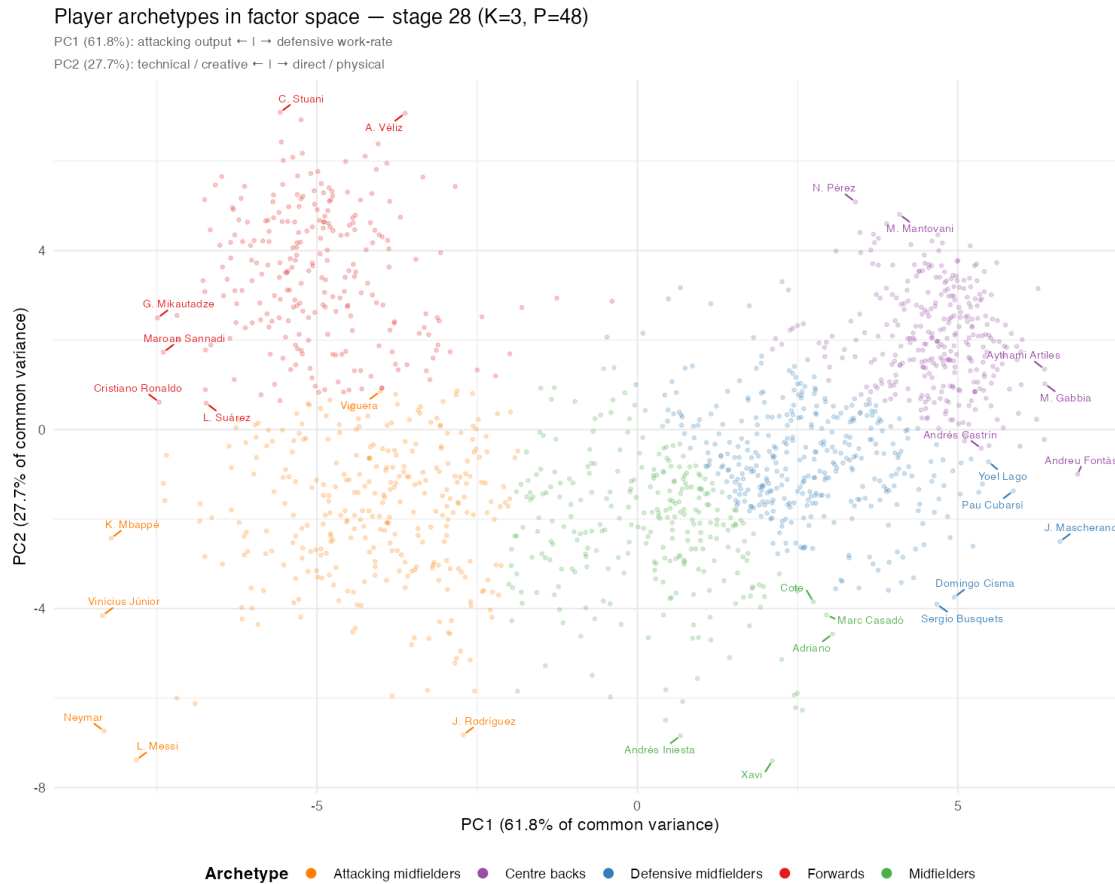


Figure 7: Player archetypes: PC1 vs PC2 for all 1,529 outfield players, coloured by k-means cluster (k=5). PC1 negative = high attacking output; PC1 positive = high defensive work-rate. PC2 negative = technical/creative style; PC2 positive = direct/physical. Extreme players labelled per cluster. Messi/Neymar/Mbappé appear in ‘Attacking midfielders’ (high attacking, high technical creativity), distinct from pure goal-scorers (Isak, Stuari, Dovbyk) in ‘Forwards’.

**Stage 28 — PC1 archetype players (61.8% of shared variance)**

**Highest PC1**

1. Andreu Fontàs 6.869 (4s)
2. J. Mascherano 6.592 (4s)
3. M. Gabbia 6.357 (1s)
4. Aythami Artilles 6.354 (2s)
5. Zé Castro 6.349 (2s)
6. Antonio Amaya 6.251 (2s)
7. Gerard Piqué 6.228 (8s)
8. A. Ba 6.003 (2s)
9. Z. Feddal 5.988 (5s)
10. Victor Ruiz 5.983 (8s)
11. S. Umtiti 5.969 (5s)
12. W. Hoedt 5.945 (1s)
13. G. Cabral 5.924 (5s)
14. NA 5.919 (1s)
15. Pau Cubarsi 5.866 (3s)

**Lowest PC1**

1. Vinicius Júnior -8.346 (8s)
2. Neymar -8.322 (3s)
3. K. Mbappé -8.213 (2s)
4. L. Messi -7.818 (7s)
5. G. Mikautadze -7.488 (1s)
6. Cristiano Ronaldo -7.461 (4s)
7. Maroan Sannadi -7.398 (1s)
8. N. Amrabat -7.397 (3s)
9. Alfon González -7.373 (1s)
10. M. Depay -7.346 (2s)
11. Lamine Yamal -7.187 (3s)
12. A. Isak -7.183 (3s)
13. K. Toko-Ekambi -7.155 (2s)
14. O. Dembélé -6.902 (5s)
15. M. Cho -6.828 (1s)

Figure 8: Top and bottom 15 players on PC1 (Offensive volume dimension). Score = posterior-mean projection; seasons = number of La Liga seasons observed.

**Stage 28 — PC2 archetype players (27.7% of shared variance)**

| Highest PC2                     | Lowest PC2                        |
|---------------------------------|-----------------------------------|
| 1. C. Sturani 7.091 (6s)        | 1. Xavi -7.401 (1s)               |
| 2. A. Véliz 7.069 (1s)          | 2. L. Messi -7.382 (7s)           |
| 3. Chris Ramos 6.922 (2s)       | 3. Andrés Iniesta -6.844 (4s)     |
| 4. A. Budimir 6.427 (7s)        | 4. J. Rodríguez -6.822 (3s)       |
| 5. Manucho Gonçalves 6.385 (2s) | 5. Neymar -6.737 (3s)             |
| 6. E. Boateng 6.175 (2s)        | 6. É. Banega -6.495 (5s)          |
| 7. S. Okazaki 6.108 (1s)        | 7. Dani Alves -6.270 (3s)         |
| 8. Y. En-Nesyri 6.087 (7s)      | 8. T. Kroos -6.214 (10s)          |
| 9. Juanmi Latasa 6.015 (3s)     | 9. O. Dembélé -6.124 (5s)         |
| 10. B. Yıldırım 5.990 (1s)      | 10. Marcelo -6.078 (8s)           |
| 11. Toché 5.951 (1s)            | 11. Lamine Yamal -6.000 (3s)      |
| 12. G. Carrillo 5.816 (4s)      | 12. Pedri -5.977 (6s)             |
| 13. Jon Bautista 5.767 (1s)     | 13. Jonathan Viera -5.953 (4s)    |
| 14. S. Weissman 5.724 (2s)      | 14. T. Gayoso Roberto -5.932 (2s) |
| 15. Diego García 5.669 (1s)     | 15. M. Pjanić -5.893 (1s)         |

Figure 9: Top and bottom 15 players on PC2 (Width / creativity dimension).

**Stage 28 — PC3 archetype players (10.5% of shared variance)**

**Highest PC3**

1. Iván Alejo 5.004 (5s)
2. Juan Iglesias 4.550 (6s)
3. O. Vranješ 4.456 (1s)
4. Houboulang Mendes 4.203 (1s)
5. A. Aznou 4.099 (1s)
6. Miguel Llambrich 3.973 (1s)
7. Rubén Sánchez 3.970 (1s)
8. Davinchi 3.851 (1s)
9. Quini 3.844 (5s)
10. L. Blanco 3.770 (1s)
11. Pacha 3.727 (5s)
12. Jesús Vázquez 3.569 (4s)
13. S. Dubarbier 3.558 (1s)
14. Pedro Porro 3.546 (2s)
15. Aitor Bunuel 3.536 (1s)

**Lowest PC3**

1. L. Messi -6.066 (7s)
2. Xavi -5.822 (1s)
3. M. Pjanić -5.464 (1s)
4. Cristiano Ronaldo -4.920 (4s)
5. K. Benzema -4.523 (9s)
6. T. Kroos -4.131 (10s)
7. Arthur -3.959 (3s)
8. Andrés Iniesta -3.948 (4s)
9. I. Gündoğan -3.928 (1s)
10. Paulinho -3.909 (1s)
11. Pedro -3.867 (1s)
12. Pau Cubarsí -3.836 (3s)
13. A. Christensen -3.721 (3s)
14. S. Nasri -3.703 (1s)
15. Philippe Coutinho -3.657 (4s)

Figure 10: Top and bottom 15 players on PC3 (Build-up / distribution dimension).

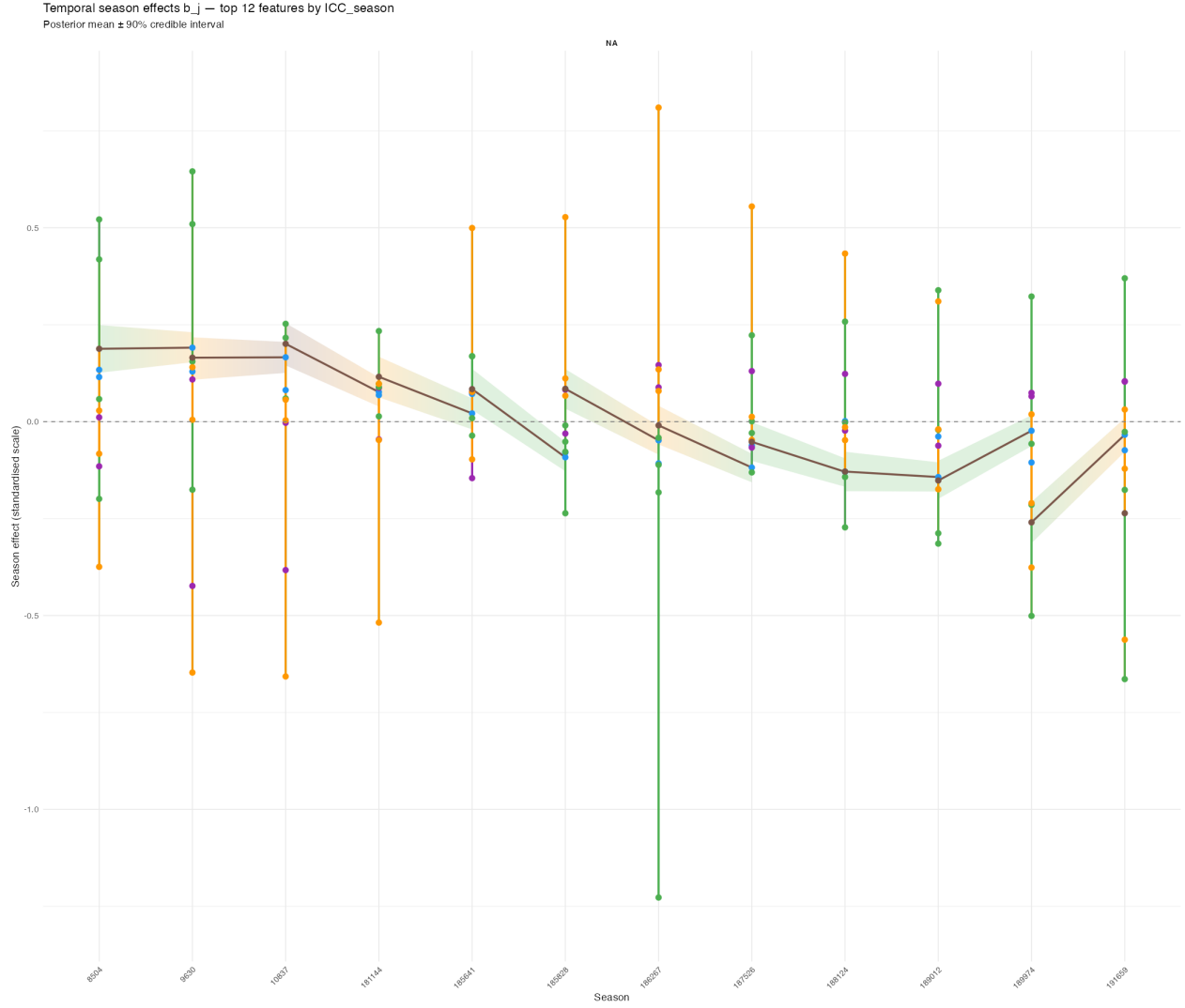


Figure 11: Posterior season effects  $\hat{b}_{j,p}$  (90% credible bands) for all 48 features, grouped by football category (stage 28,  $K=3$ ,  $N = 4,586$ ). Season 1 = 2011/12, Season 12 = 2022/23. Most ICC\_season < 5%; bands are wide relative to trends — interpret conservatively.

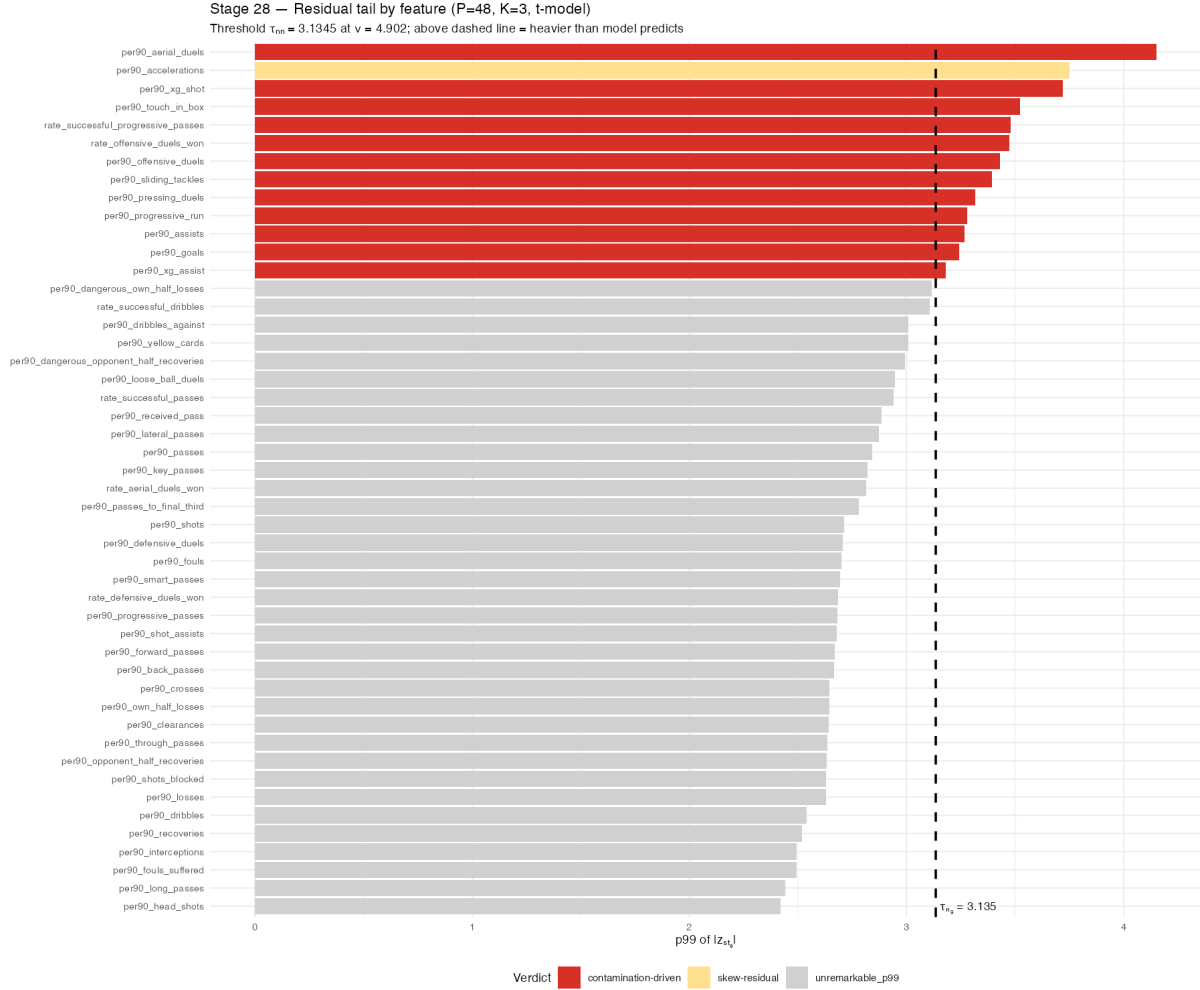


Figure 12: p99 of  $|z_{np}|$  for each of the 48 features, coloured by verdict. Dashed line =  $\tau_{99} = 3.135$  (model's own 99th-pct threshold at  $\hat{v} = 4.902$ ). Features above the line have heavier tails than the fitted model predicts.

- **Smart-passes extreme (Reyes 2012/13).** This was flagged at  $z \approx 14.4\sigma$  in the pre-rebuild study. After `log1p` transform on `per90_smart_passes`, the distribution is compressed. The same observation may be significantly smaller now.

=== Old Task A vs Stage 28 comparison ===

Old tau99 (nu=2.95): 3.3709

New tau99 (nu=4.9023): 3.1345

Lighter tails (higher nu) → lower threshold → model more surprised by extremes

rate\_defensive\_duels\_won frac > tau99: 0.0037 (old: GK-contaminated, should be lower)

Reyes 2013/14 per90\_smart\_passes: not found in rebuilt data (old: 14.4)

Total (n,p) cells > tau99: 1702 / 220128 (0.77%)

Expected under t(nu): ~2.65%

## 8 Explained Variability (ICC)

For each feature  $p$ , the intra-class correlation (ICC) decomposes total variance:

$$\text{ICC}_{\text{player}} = \frac{\Sigma_{a,pp}}{\Sigma_{a,pp} + \Sigma_{b,pp} + \Sigma_{e,pp}}, \quad \text{where} \quad \Sigma_{a,pp} = \Lambda_p \Lambda_p^\top + \psi_{a,p}^2$$

Table 13: Top 10 features by player ICC (K=2 Normal model, rebuilt P=48 data). Player ICC = proportion of total variance attributable to stable between-player differences. High values indicate genuine skill signal.

| Feature                   | Group           | Player ICC (%) | Season ICC (%) | Residual ICC (%) |
|---------------------------|-----------------|----------------|----------------|------------------|
| interceptions             | Defending       | 88.7           | 1.1            | 10.2             |
| dangerous own half losses | Defending       | 87.1           | 0.1            | 12.8             |
| loose ball duels          | Dueling         | 87.1           | 5.2            | 7.8              |
| shots                     | Attacking       | 85.7           | 1.0            | 13.3             |
|                           | output          |                |                |                  |
| back passes               | Passing volume  | 85.4           | 1.0            | 13.6             |
| lateral passes            | Passing volume  | 85.0           | 0.4            | 14.5             |
| successful passes         | Passing quality | 84.4           | 0.8            | 14.8             |
| progressive passes        | Passing volume  | 84.2           | 0.4            | 15.4             |
| offensive duels           | Dueling         | 84.0           | 2.9            | 13.2             |
| long passes               | Passing volume  | 82.0           | 0.8            | 17.2             |

*Note:* ICC values are from the K=2 Normal model (stage 10/11) on rebuilt data. ICC from the K=3 t-model (stage 28) pending stages 11–16 rerun.

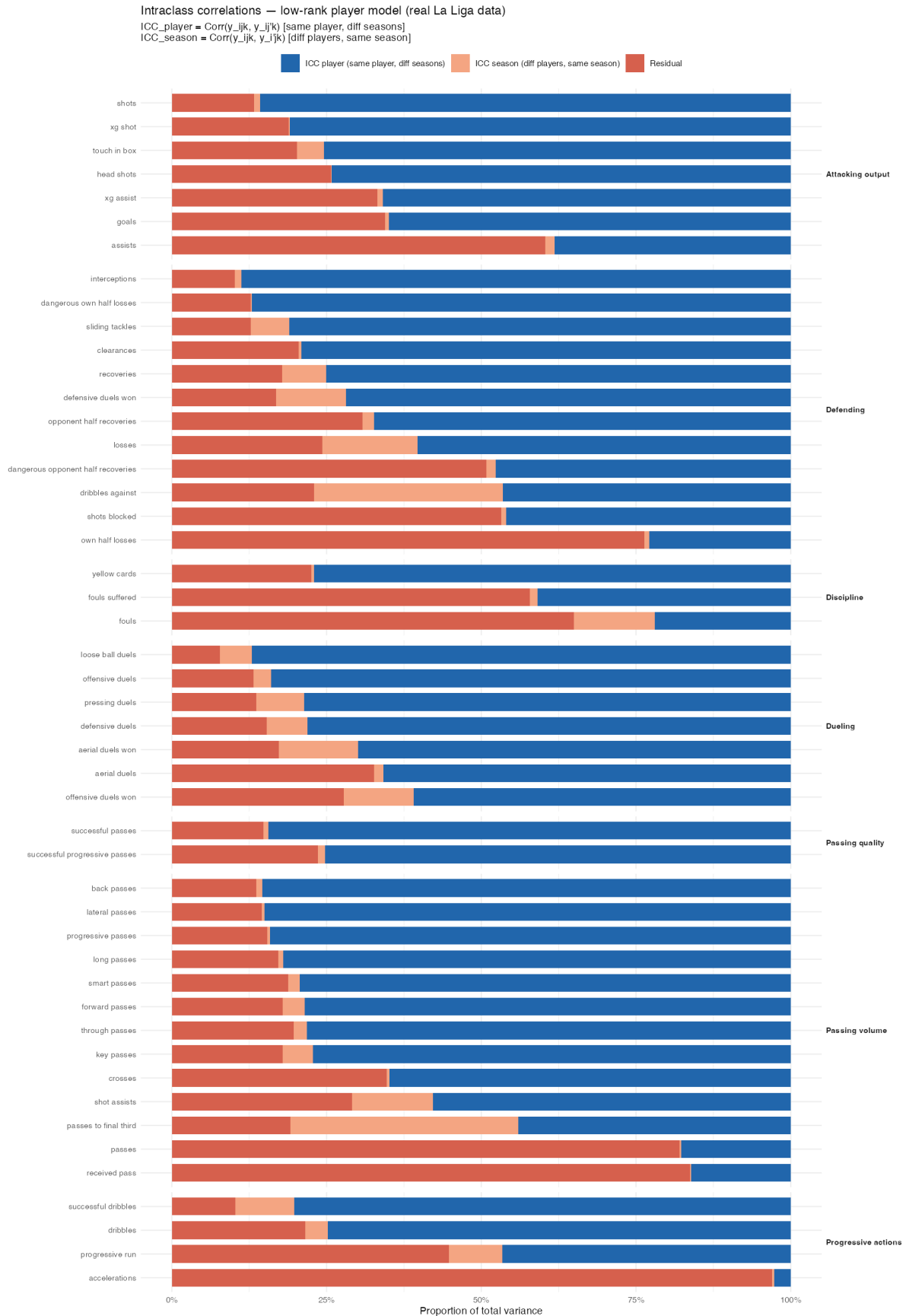


Figure 13: ICC decomposition for all 48 features (K=2 Normal model, rebuilt data P=48), ranked by player ICC. Blue = player (stable skill), orange = season, grey = residual.



## 9 Model Comparison Table

Table 14: All model comparisons to date. \* Pathfinder values are variational approximations; direction reliable, absolute magnitudes approximate. \*\* PSIS-LOO unreliable at 99.4% Pareto- $k > 0.5$ ; direction consistent with Pathfinder.

| Comparison                                       | $\Delta\text{ELPD}$ | SE    | SE ratio | Source                                         |
|--------------------------------------------------|---------------------|-------|----------|------------------------------------------------|
| Low-rank $K = 2$ vs diagonal $K = 0$             | +3,724              | 187   | 20       | Stage 20 PSIS-LOO ( $P = 53$ , $N = 4944$ )    |
| $t$ -errors vs Gaussian ( $K = 2$ , pre-rebuild) | +11,668             | 554   | 21       | Stage 18b PSIS-LOO ( $P = 53$ , $N = 4944$ )   |
| Low-rank $K = 3$ vs $K = 2$ (Pathfinder)         | +12,125*            | —     | —        | Pathfinder screening ( $P = 48$ , $N = 4586$ ) |
| Low-rank $K = 3$ vs $K = 2$ (PSIS-LOO)           | +1,244**            | 1,271 | 0.98     | Block C PSIS-LOO ( $P = 48$ , $N = 4586$ )     |
| Low-rank $K = 3$ vs diagonal $K = 0$ (LOO)       | pending             | —     | —        | Stage 20 rerun needed ( $P = 48$ )             |

**Summary:** All comparisons favour the production model ( $K=3$   $t$ -errors). The low-rank structure is real (20 SE over diagonal). Student- $t$  residuals are strongly supported (21 SE improvement pre-rebuild; structurally justified post-rebuild with  $\hat{\nu} = 4.90 \neq \infty$ ). Rank  $K=3$  is preferred over  $K=2$  in both Pathfinder screening and PSIS-LOO direction.

## 10 Posterior Predictive Checks

## 11 Open Questions and Next Steps

### 11.1 Production model status

| Item               | Value                                              | Status   |
|--------------------|----------------------------------------------------|----------|
| Model              | Stage 28: $K=3$ , $P=48$ , $N=4,586$ , $t$ -errors | Complete |
| $\hat{\nu}$        | 4.902 (95% CI [4.79, 5.01])                        | Complete |
| Lambda_a mixing    | ESS=237, Rhat=1.005 (PASS)                         | Complete |
| LOO $K=3$ vs $K=2$ | +1,244 $\pm$ 1,271 (direction: $K=3$ preferred)    | Complete |

| Item                                   | Value                                            | Status   |
|----------------------------------------|--------------------------------------------------|----------|
| Marginalized Normal<br>( $I = 1,000$ ) | Coverage $\geq 85\%$ (PASS)                      | Complete |
| Outlier study (stage 29)               | All 48 features, 1,529 players,<br>top 200 cells | Complete |

## 11.2 Pending analyses

1. **Stage 20 rerun.** Diagonal baseline needs refitting on rebuilt data ( $P = 48$ ,  $N = 4,586$ ,  $K=0$  stage 03). Currently running (Session 9 Block D). Once complete, the low-rank vs diagonal comparison becomes final on the rebuilt data.
2. **Stages 11–16 on stage 28.** Full ICC, loading analysis, player/season profiles for the  $K=3$  production model. Currently these sections use the  $K=2$  Normal model as proxy.
3. **Factor interpretation.** PC1/PC2/PC3 labels are preliminary. Draws-level PCA of  $\Lambda\Lambda^\top$  gives posterior intervals on the eigenvectors, enabling credible-interval loading plots.
4. **Structured group prior.** A prior that encourages loadings to be sparse within feature groups could improve interpretability and regularisation.

## 11.3 Longer-term modelling

5. **Team-season effects.** A team-season effect  $c_{t[n]} \in \mathbb{R}^P$  would separate manager and tactical system effects from stable individual traits. Required: a player-team mapping, which is available in the raw data.
6. **Temporal structure.** Season effects  $b_j$  are currently exchangeable. A Gaussian process or random walk over time would capture league-wide temporal trends with appropriate smoothing.
7. **Two-component mixture (deprioritised).** At  $\hat{\nu} = 4.90$ , the tail is moderate and well-characterised by Student- $t$ . The mixture model (outlier fraction vs smooth heavy tail) is theoretically appealing but practically unnecessary given the current  $\hat{\nu}$ .

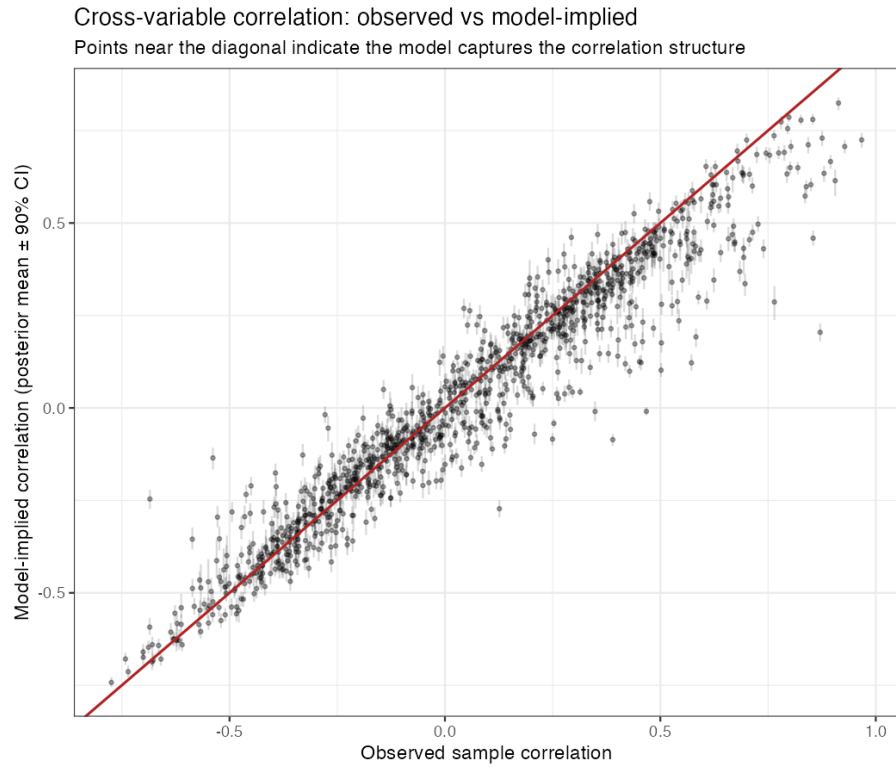


Figure 14: PPC: observed vs predicted feature-pair correlations (K=2 Normal model, P=48 rebuilt data). High correlation ( $r \approx 0.98$ ) confirms the model captures cross-feature covariance structure. PPC for stage 28 K=3 t-model pending stage 19 rerun.